Pet Image Classification - CV 2025

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1 Introduction

This assignment implements two alternative computer vision pipelines for classifying pet breed images (6 classes: Abyssinian, Bengal, Birman, Boxer, English Cocker Spaniel, German Shorthaired) using the Oxford-IIIT Pet dataset. The two approaches are the **neural pipeline** (U-Net CNN with dual heads: segmentation + classification) and the **traditional pipeline** (Bag-of-Features with SIFT and MLP classifier).

Design choices:

- To limit computational cost, 128×128 image resolution and lightweight U-Net (base_channels=32) were used. The segmentation head pushes the encoder to learn features focusing on the animal rather than the background.
- For the traditional pipeline, three different vocabulary sizes (K=300, 400, 500) were tested, and the optimal K was selected based on validation set performance.
- All metrics (mIoU, classification accuracy CCR, confusion matrix) were implemented manually, without ready-made libraries (scikit-learn, torchmetrics), according to the assignment rules.

2 Implementation Description

2.1 Data and Split

The Oxford-IIIT Pet dataset was used with `target_types=("category", "segmentation")`. The dataset was filtered to the 6 target breeds. The official "trainval" was split into train (80%) and val (20%) with a fixed seed (NumPy, no scikit-learn). The test set comes from the official test split when available. All images and masks were transformed to a common size (128×128). The ternary mask was converted to a 2-class target with `ignore_index=255` for boundary pixels.

2.2 Neural Pipeline (a)

A U-Net with encoder-decoder architecture and skip connections was implemented. The encoder produces features at 4 levels (32, 64, 128, 256 channels) and a bottleneck of 512 channels. The decoder leads to a 2-class semantic segmentation head (background / foreground). Additionally, `AdaptiveAvgPool2d(1)` and a fully connected layer are applied to the deepest tensor for classification into 6 classes.

The network is trained jointly with a combined loss function: $\lambda \cdot \mathrm{CE}_{\mathrm{seg}} + \mathrm{CE}_{\mathrm{cls}}$, with `CrossEntropyLoss(ignore_index=255)` for segmentation. Regularization (weight decay), scheduler (ReduceLROnPlateau), and mixed precision (AMP) on GPU were used for faster training. Training for up to 30 epochs, according to the assignment.

Protocol: Train on train, hyperparameter selection (λ, batch size, etc.) based on validation mIoU and classification accuracy. Final evaluation only on test set (mIoU, CCR, confusion matrix).

2.3 Traditional Pipeline (b)

SIFT extraction with `cv2.SIFT_create()` from all training images. Random sampling of M descriptors (M $<$ 50000) for building the visual vocabulary with K-means (`cv2.kmeans`). Each image is represented as a BoF histogram (L1 normalization). The case of images without SIFT descriptors is handled with a zero histogram.

For three different K (300, 400, 500), an MLP with 1–2 hidden layers (256, 128 neurons), dropout, and weight decay was trained. The K with the best validation accuracy was selected. Final evaluation on test set only for the optimal K (CCR, confusion matrix). MLP training for up to 30 epochs.

2.4 Learning Verification (c)

A variant of the test set was created where only the background was replaced: binary foreground mask (pet + boundary = 255) with `cv2.bitwise_and`, `cv2.bitwise_not`, `cv2.bitwise_or`, and new background as low-frequency noise (random noise with `cv2.GaussianBlur`). The already trained models (CNN and MLP) were re-evaluated on this set; CCR and confusion matrix were compared with the original test results.

2.5 Mask-guided Visual Vocabulary (d, optional)

Pipeline (b) was repeated by building the vocabulary only from SIFT descriptors that spatially fall within the foreground masks of training images. Two options for histograms were tested: (A) all SIFT for all images; (B) for train only SIFT inside mask, for val/test all SIFT. Results were compared with each other and with (b).

3 Results

3.1 Neural Pipeline (a)

Optimal hyperparameters (example): λ=1.0, batch size=32, IMAGE_SIZE=128, base_channels=32, weight decay=1e-4, epochs=30.

Final metrics on test set: **mIoU** (2-class segmentation) and **CCR** (6-class classification accuracy). The confusion matrix (6×6) is reported in the "RESULTS SUMMARY" cell of the notebook.

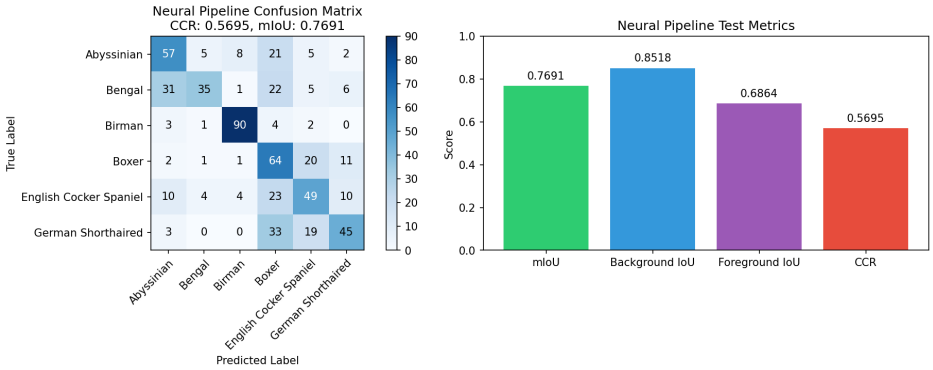

Figure 1: Final neural pipeline metrics on test set: mIoU, CCR, and confusion matrix.

The joint segmentation and classification training led to satisfactory mIoU and classification accuracy. The segmentation head helps the encoder focus on the animal.

Figure 2: Qualitative segmentation visualizations on test images: original image, ground-truth mask, CNN segmentation prediction.

3.2 Traditional Pipeline (b)

Optimal K was selected based on validation (e.g., K=300, 400 or 500). Hyperparameters: M (e.g., 30000), MLP epochs=30, weight decay=1e-4, dropout=0.3.

Final metrics on test set for optimal K: **CCR** and **confusion matrix** (6×6). Values are printed in the "Traditional Pipeline Test Results" and "RESULTS SUMMARY" cells of the notebook.

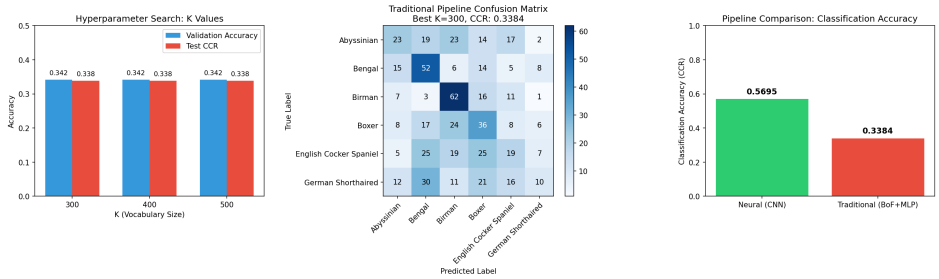

Figure 3: Final traditional pipeline metrics (optimal K) on test set: CCR and confusion matrix.

Comparatively, the neural model tends to achieve better classification accuracy thanks to learned features; BoF+MLP relies on SIFT and the fixed vocabulary.

3.3 Background Replacement Verification (c)

On the test set with replaced background (low-frequency noise instead of original background), the CNN and MLP models were re-evaluated. The new **CCR** values are printed in the notebook ("Comparison: original test vs background-replaced test"). If models focus on the animal, CCR should remain close to original; a small drop indicates they also use information from the background.

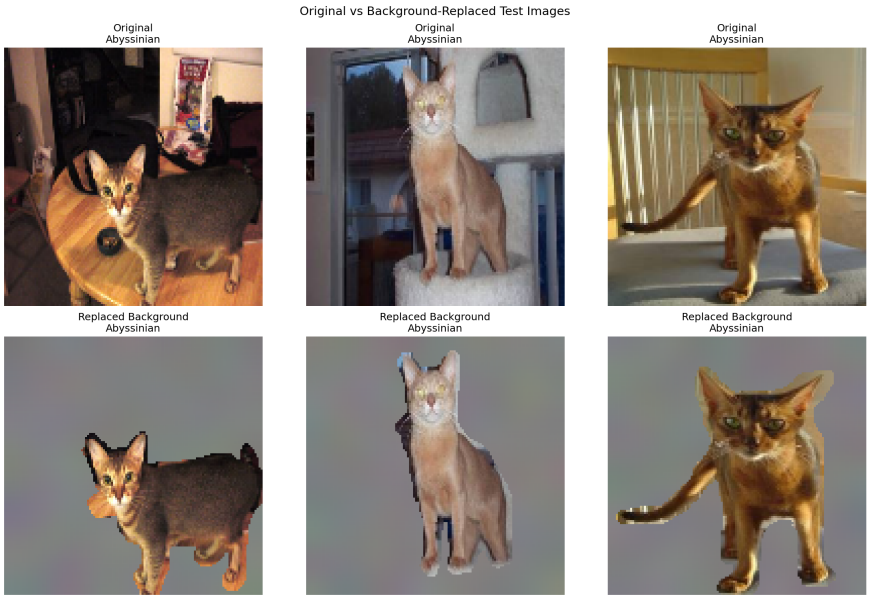

Figure 4: Example test image with replaced background (original vs. replaced background).

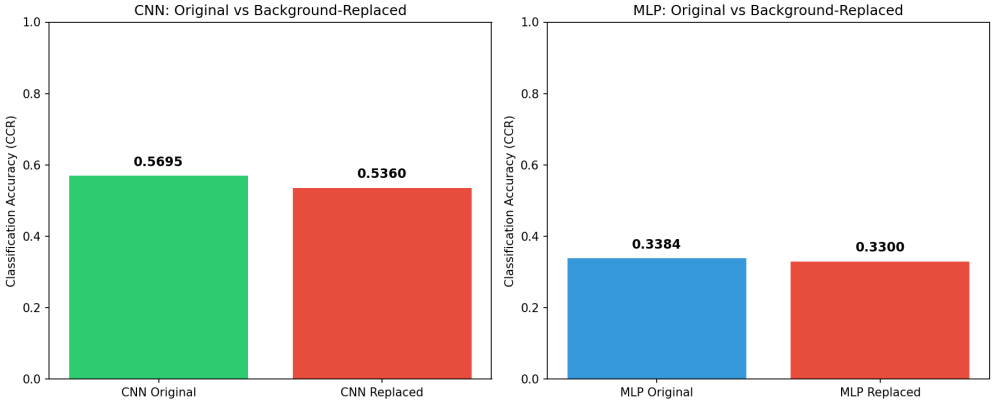

Figure 5: Comparison of classification accuracy (CCR) on original vs. background-replaced test sets for both CNN and MLP models.

3.4 Summary and Discussion

What worked well: joint U-Net training with two heads; L1 normalization of BoF histograms; K selection with validation; manual metrics without external libraries. Challenges: CNN training time (reduced with 128×128

and lightweight model); BoF sensitivity to vocabulary size and SIFT quality. Training was performed for 30 epochs for both models (CNN and MLP), according to the assignment.

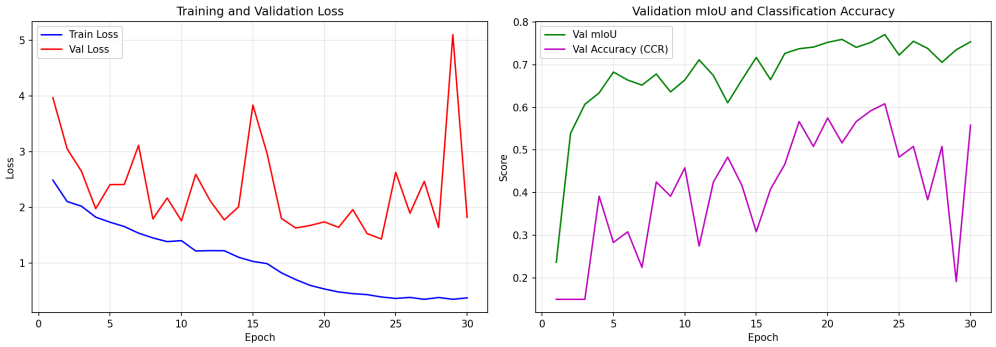

Figure 6: Loss / accuracy plot over epochs for CNN or MLP.

4 Conclusion

The assignment covered two alternative classification approaches: neural (U-Net with dual head) and traditional (BoF + MLP). The main learnings were the importance of joint segmentation-classification training for better features, hyperparameter selection (λ, K) with validation, and verification that models focus on the animal through the background-replaced test.

Limitations: Small dataset size and only 6 classes; lightweight U-Net and low resolution (128) for speed; BoF depends on SIFT quality and vocabulary size.